Deferral-Aware Digital Twin

The complete guide: what it is, how every part works from the first click to the final answer, how it was evaluated, and how to explain it in your review.

> **How to use this guide.** Part A explains the idea. Part B follows the app from installation to the first answer. Part C opens up the backend: memory, the deferral gate, the language model, learning, Online and Expert mode, languages, voice, lip-sync and the character. Part D covers the results. Part E is a file map, the tech stack, limitations, and the questions examiners are likely to ask, with answers.

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1 · The problem

A **digital twin of a person** is software that answers questions the way that person would: their opinions, habits, choices and way of talking. The obvious way to build one today is to give a chatbot (ChatGPT, Gemini…) some facts about the person and ask it to role-play.

That has a serious flaw: **language models never say "I don't know"**. Ask a role-playing chatbot "What is your passport number?" or "What did you eat on 3 March?" and it confidently invents an answer. For a twin that is meant to represent a real person, a confident invention is worse than no answer: it puts words in the person's mouth.

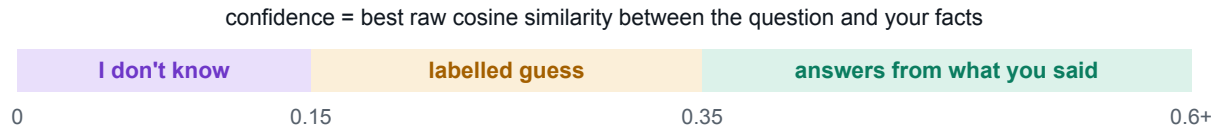
- Chatbots **hallucinate**: they fill gaps with plausible fiction.
- They give **no evidence**: you can't see which of the person's real statements an answer came from.
- Making them more personal usually means **fine-tuning** (training) on the person's data, which is slow, needs GPUs, and bakes private data into model weights.
- Most run in the **cloud**, so private answers leave the device.

Research question

> Can we build a digital twin that answers like the person when it has evidence, **says so when it is only guessing**, and **refuses ("I don't know") when nothing the person said is related**, while learning from corrections instantly, without any model training, on an ordinary laptop?

2 · The solution in one page

The twin is built on **retrieval**, not training. Every interview answer becomes a **fact** in a searchable memory. When a question comes in, the twin finds the facts closest in meaning and measures **how close** the best one is. That number, the **confidence**, decides what the twin is allowed to do:



Confidence	Tier	What the twin does	Shown as
below 0.15	hard_defer	Refuses. No language model is even called, so it cannot invent anything.	"I don't know" (purple)
0.15 to 0.35	soft_answer	Reasons from the person's profile and the weak matches, and labels it as a guess.	"You never answered this · reasoning from your profile" (amber)
0.35 and above	confident_answer	Answers from what the person actually said, and shows that answer as evidence.	"You said this" (green)

What makes it novel

- **Deferral as the core idea**: three honest tiers decided by a measurable number, not by the model's own opinion.
- **Honest confidence**: the gate always uses the *raw* similarity. Facts the person stated get a small ranking boost, but the boost never reaches the gate, so adding facts can't fake confidence (there is a test for this).
- **Learning without training**: a correction is added to memory and works on the very next question. Nothing is fine-tuned.
- **Evidence for every answer**: "Why this answer" shows which real answers were used, with a bar for each.
- **Runs on a laptop**: Phi-3 on Apple's GPU, voice cloning and lip-sync locally. Online mode (Gemini) is optional.
- **Multimodal**: the twin speaks in the person's cloned voice (English and Hindi) through a talking avatar from one photo, understands speech (Whisper) in English, Hindi and Telugu, and has an animated landing character.
- **Measured**: tested on 100 people from a public benchmark and on the real subject's own held-out answers.

3 · The big picture

The app is a normal Mac application. Inside, it is a small website served by a local Python server, shown in a native window. Nothing listens on the network: the server only accepts connections from the same Mac (127.0.0.1).

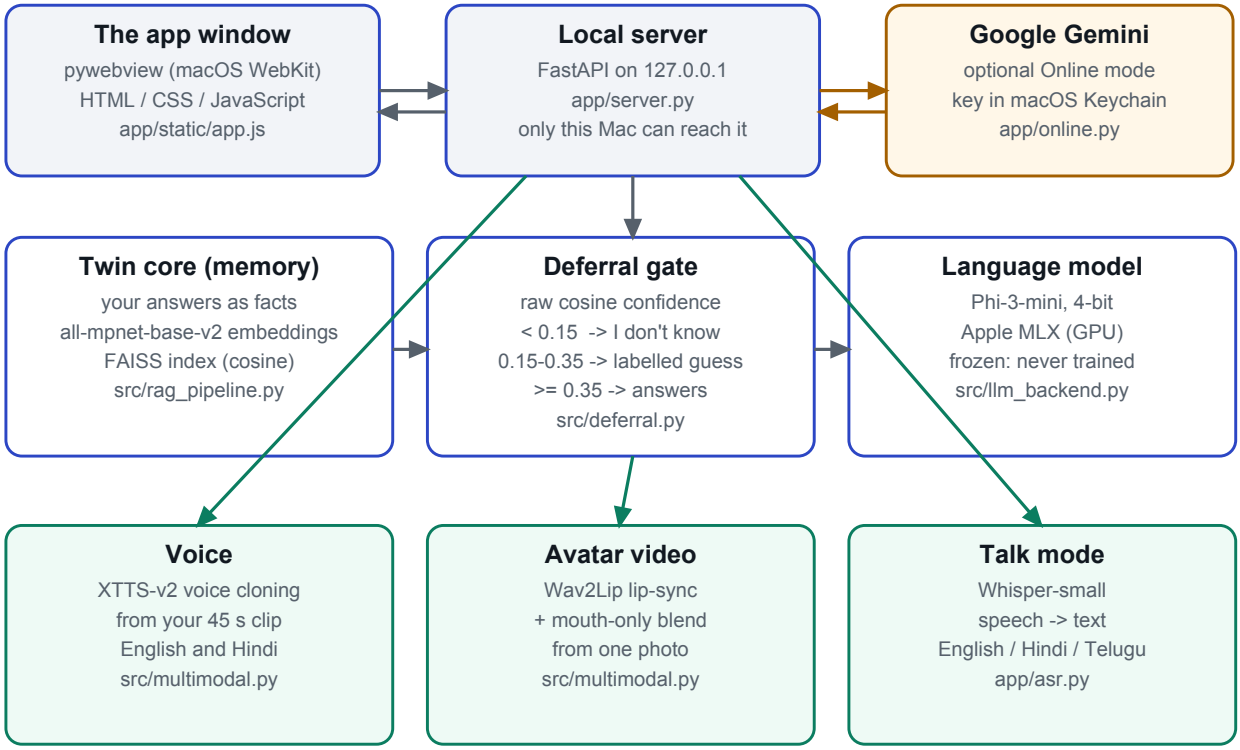


Figure 1. The parts of the system. Green boxes are optional media features; amber is the optional cloud part.

The three layers, in plain words

- **Front end** (what you see): HTML, CSS and JavaScript in `app/static/`. It draws the pages, the animated character, the chat, the charts, and sends requests to the server.
- **Server** (the coordinator): `app/server.py`, built with FastAPI. It receives each request ("ask this", "save my photo", "make a video"), calls the right component and returns JSON.
- **Twin core** (the brain): memory and retrieval (`rag_pipeline.py`), the deferral gate and prompt building (`deferral.py`), the person's profile (`persona.py`), and the language model (`llm_backend.py`).

Heavy models run in separate Python environments (main, Wav2Lip, GFPGAN) so their libraries don't conflict. The server starts them as separate processes when needed.

4 · Installation and first-run setup

DeferralTwin.app is small (about 200 MB): it contains a Python runtime, the launcher and the project code, but not the AI models. On first launch the **bootstrapper** (`bootstrap/engine.py`) sets everything up in `~/Library/Application Support/DeferralTwin`:

Step	What happens
Environments	uv creates isolated Python environments from locked dependency lists (bootstrap/locks/*.lock), so every Mac gets identical versions.
Models	Downloads Phi-3-mini 4-bit (MLX), the all-mpnet-base-v2 embedder, XTTS-v2 (voice), Wav2Lip + face detector, GFPGAN, the English→Hindi translator. Resumable downloads.
ffmpeg	Links the static ffmpeg build that ships inside a Python wheel (no Homebrew needed).
Self-test	A benchmark measures this Mac: model load time, answer time, voice and lip-sync speed. The app uses these numbers for time estimates.
Launch	The launcher (bootstrap/launcher.py) starts the local server and opens a pywebview window (macOS WebKit). Camera and microphone permission texts come from the app's Info.plist.

Talk mode (Whisper, about 970 MB) and the Hindi→English translator are downloaded once on first use.

Where your data lives

```
~/Library/Application Support/DeferralTwin/
  twin/phase2_answers.json      your interview answers
  twin/ref.wav, photo_fixed.jpg your voice sample and photo
  twin/benchmark/phase2/phase2_me/
     train.jsonl / test.jsonl  facts the twin knows / answers held back for testing
     faiss_demo.bin            the memory index
     feedback_demo.jsonl       your corrections   notes.jsonl  things you told it
     profile.json              your computed profile and writing style
  twin/online_profile.json     the Gemini-written profile (Online mode only)
  twin/work_profile.json       your job and skills (Expert mode)
  models/  envs/  cache/       downloaded models, Python environments, cached
audio/video
```

5 · Consent and the interview

Consent

Before anything is recorded, the person agrees to how their data is used. Voice cloning and the avatar are enabled **only** for the person who consented (a hard rule of the project). Online mode has its own consent line, because it sends text to Google.

The interview: 109 questions in 8 sections

Section	Example	Answer type	Why it's there
About you (12)	Which city do you call home? What are you studying?	free text	basic facts, background
Everyday preferences (13)	Tea or coffee? Morning person or night owl?	short choice	tastes and habits
What would you do? (16)	A friend cancels plans last minute. What do you do?	2–3 sentences	behaviour in situations
What you believe (14)	Learning by building beats learning by reading.	1–5 agreement (Likert)	values and opinions
How you tend to be (20)	I start conversations with new people easily.	1–5 agreement	personality (Big Five)
Either or (12)	Stable work or risky work with a big upside?	one of two	trade-offs
In your own words (10)	Describe a perfect day.	longer text	voice and writing style
Quick re-check (12)	Reworded repeats of earlier questions	same as original	measures your own consistency (the human ceiling)

Answers are autosaved as you type. Then the person records about 45 seconds of a read-aloud paragraph (voice) and one photo; both are checked immediately (too short, clipped or too quiet audio is rejected; the photo must show one whole face with the mouth visible).

6 · Building the twin (no training)

Pressing **Build my twin** runs `load_phase2.py` and the build steps. Nothing is trained. It is data preparation plus an index:

Step	What it does	Result
1 Split	A seeded, stratified 25% of answers is **held out**: the twin never sees them. (Stratified = every section is represented.)	train.jsonl (what it knows), test.jsonl (18 secret answers for testing)
2 Facts	Each remaining answer becomes a fact written as "Q: question A: answer".	79 facts for Hardhik
3 Embed	Each fact is turned into a 768-number vector by **all-mpnet-base-v2**, a sentence-embedding model. Texts with similar meaning get vectors pointing in similar directions.	79 vectors
4 Index	The vectors go into a **FAISS** IndexFlatIP index. Because the vectors are normalised, inner product = cosine similarity.	faiss_demo.bin
5 Profile	persona.py computes a profile: how you rate statements, your either-or habits, Big Five, and a **style card** (typical answer length, lowercase or not, example answers).	profile.json
6 Baseline + consent	A simple baseline for comparison and the consent record are saved.	baseline.jsonl, consent.json

> **Why retrieval instead of training?** Training (fine-tuning) changes the model's weights: it is slow, needs a strong GPU, can't be undone easily, and the model may still invent things. Retrieval keeps the model frozen and gives it the person's own words at question time. It is instant to update, transparent (we can show the evidence), and it gives us a number (similarity) to decide when to refuse.

7 · What happens when you ask a question

Every question goes through the same pipeline (`_ask` in app/server.py, then deferral.py):



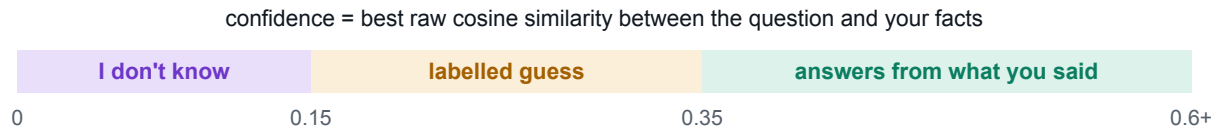
Figure 2. The answering pipeline, step by step.

Retrieval and cosine similarity, explained

The question is embedded into a vector with the same model as the facts. FAISS compares it with all fact vectors and returns the top 8. The comparison is **cosine similarity**: the cosine of the angle between two vectors. 1 means the same meaning, 0 means unrelated. Example scores for Hardhik's twin:

Question	Closest fact	Score	Tier
Tea or coffee?	Q: Tea or coffee? A: Both	0.8	confident
What do you do in your free time?	Q: What do you do to relax? …	0.55	confident
Do you know linear algebra?	(no close fact)	0.19	labelled guess
What is your passport number?	(nothing related)	below 0.15	I don't know

8 · The deferral gate, in detail



The gate is three lines of logic, but it is the heart of the project:

```
results, confidence = memory.retrieve(question)    # confidence = best RAW cosine
if confidence < 0.15:   tier = 'hard_defer'         # refuse, no model call
elif confidence < 0.35: tier = 'soft_answer'        # guess from profile, labelled
else:                   tier = 'confident_answer'   # answer from the person's words
```

Why "raw" similarity matters

Facts the person stated directly (corrections and notes) get a **+0.05 ranking boost**, so that "I moved to Chennai" outranks the old interview answer "I live in Tirupati". But the boost is used **only for ordering**. The confidence that the gate sees is the raw cosine. Otherwise, simply adding more notes would push weak matches over the thresholds and fake confidence. A dedicated test (tests/test_gate_and_media.py) keeps this honest.

Where the thresholds come from

0.15 and 0.35 were chosen on the public Twin-2K-500 benchmark (Phase 1) by looking at how answer quality changes with similarity (the risk–coverage curve, section 18). They are kept fixed; all later features (Online mode, Expert mode, languages) leave them unchanged.

What each tier does after the gate

- **hard_defer**: returns a fixed "I don't know" sentence. The language model is not called at all, so it is impossible to hallucinate.
- **soft_answer**: the prompt includes the person's profile and the weak matches, and asks for an in-character guess. The answer card says "You never answered this · reasoning from your profile".
- **confident_answer**: the prompt includes the closest real answers, and the model answers from them. The card shows the matching question and answer as evidence.
- **Format follows the question**: a statement to rate gets "Agree (4 of 5)"; an either-or question gets one choice; everything else gets free text, at the person's usual length.

9 · The prompt and the persona

For tiers that call the model, deferral.py builds a **system prompt**. Its pieces:

- **Identity rules**: speak as the person in first person; never say "I am an AI"; if asked what you are, "I'm X's digital twin".
- **Style card**: "usually about N words, mostly lowercase, direct and casual" plus a few of the person's real answers as examples, computed from the interview so the twin *sounds* like them.
- **Evidence**: the retrieved facts (the person's own words).
- **Profile summary**: for guesses, how the person tends to behave.

The model is **Phi-3-mini** (3.8 billion parameters), quantised to 4 bits so it fits in about 2.4 GB, running on the Mac's GPU through **Apple MLX**. A typical answer takes about 1 second. Its output is cleaned: phrases like "As an AI…", "Great question!", and sympathy/advice openers are removed, and a sentence cut off by the length limit is trimmed.

Special questions skip memory entirely: the clock ("what time is it?"), questions about the twin ("who are you?", "what can you do?") and greetings are answered from templates, labelled "About your twin · not from your memories".

10 · Learning without training

The twin improves in four ways, and none of them changes the model:

Way	How the person does it	What happens inside
Correction	"That's not me" under an answer → type the real answer	A new fact "Q: … A: …" (source user_correction) is added to the FAISS index and logged to feedback_demo.jsonl. It outranks the old fact (+0.05 boost).
"That's me □"	Confirm a good guess	The guessed answer becomes a confirmed fact.
Notes	Teach page: one thought per line	Each line becomes its own fact (source user_note).
Gaps	Teach page lists questions it refused or guessed	Answering one adds it as a correction.
Guess me	A game: the twin locks a guess, you answer	Your answer is stored as a correction; the match score is logged.

On restart, all corrections and notes are replayed into memory, so nothing is forgotten. In Phase 1, five corrections lifted similarity on related questions from 0.35 to 0.563 (section 18). This is the "learns instantly, no training" claim, measured.

11 · Online mode (Google Gemini, optional)

Off by default. When turned on, the twin can reason better about new situations using Gemini, while the deferral gate still decides what it may answer.

How it works

- **Key**: pasted once, checked with Google, stored in the **macOS Keychain** (never in a file, log or response). Errors are scrubbed so the key can't leak.
- **Model choice**: the app asks the API which models exist and builds a chain: newest Pro → newest Flash models → Flash-Lite. On the free tier Pro has no quota, so Flash usually answers. A model out of quota is skipped for 30 min, a busy one for 1 min.
- **Profile written once**: Gemini reads the interview, corrections and notes **once** and writes a first-person profile (values, personality, habits, how you decide, how you talk). It is rewritten only when your memory changes (fingerprint check), so each question doesn't resend the whole interview.
- **Each question** sends: persona rules + date/time + the profile + the top retrieved real answers + the earlier chat turns.
- **Same gate**: below 0.15 there is no Gemini call at all; the tier label still comes from the raw confidence.
- **Fallback**: no internet, no quota, any error → the local answer, marked "offline · this mac" with the reason.
- **Voice and video always stay on the Mac.**

12 · Expert mode

A twin of a civil engineer should be able to answer "what causes cracks in walls?" even though that was never in the interview. Expert mode does this **honestly**:

- **Job and skills**: picked up from the interview answers by Gemini (Hardhik: "AI & data science student"), or typed on the Teach page with an optional CV.
- **Field subjects**: Gemini lists about 70 subjects someone in that job knows (half everyday problems, half core concepts).
- **Recognising a field question**: the question is embedded and compared with those subjects. Cosine ≥ 0.40 → field question; 0.22–0.40 → Gemini is asked a one-word yes/no; personal questions ("do you like…", "your favourite…") never count.
- **Only when your own memory has nothing confident** (raw confidence < 0.35). If you said something related, your answer wins.
- **Answer**: Gemini's domain knowledge, in your tone, labelled **"Expert knowledge · <field>"**, with a note that it isn't a personal memory.
- **Outside the field, and offline, the normal gate applies**: an AI student's twin still says "I don't know" to a concrete-grade question.

13 · Telugu, Hindi and chat memory

Asking in Telugu or Hindi (app/lang.py)

Memory and retrieval are in English, so the app works in three steps: (1) detect the script (Telugu or Devanagari Unicode ranges), (2) turn the question into one standalone English question (Gemini online; offline, a local Hindi→English Marian model), (3) run the normal pipeline and gate, then translate the answer back into the question's language **and script**. Example: a Telugu question "which is your town?" → "What is your village?" → "My hometown is Tirupati" in Telugu script. Hindi answers can be spoken in the cloned voice; Telugu is text-only because the voice model has no Telugu.

Remembering the chat

The last 6 turns are sent with each question. A short follow-up ("why?", "and on weekends?") is rewritten into a full question before retrieval, so the gate still scores a real question: "why?" after "Which city do you call home?" becomes "Why do you call Tirupati home?". Online answers also see the earlier turns. "New chat" clears the conversation, not the memory.

14 · Talk mode, voice and the avatar

Talk mode: speech → text (app/asr.py)

Press the mic, speak, press again. The recording is converted to 16 kHz audio and transcribed by **Whisper-small** (OpenAI, multilingual) on the Mac's CPU in about 1–2 s. Whisper detects the language. Its small model often writes Telugu speech in Devanagari letters; because the two Unicode blocks share one layout, the app shifts each letter to Telugu script. The text then goes through the normal pipeline, and the answer is spoken in your voice.

The cloned voice: text → speech (XTTS-v2)

XTTS-v2 (Coqui) clones a voice from a short sample: it encodes the 28-second `ref.wav` into a speaker representation and generates new speech in that voice. Text is spoken sentence by sentence (splitting on ".", "?", "!" and the Hindi danda "□"), with tuned sampling (top-p 0.7, top-k 30), trimmed silences and even loudness. Hindi uses Gemini's translation when online, else the local Marian model.

The talking avatar (Wav2Lip + mouth-only blend)

Wav2Lip takes the photo and the voice audio and generates, for every video frame, a mouth shape matching the sound. It regenerates the whole lower face at only 96×96 pixels, which makes the face blurry and leaves a box edge. The app then **keeps the original photo everywhere** and blends in only the mouth and jaw through a soft oval, sharpened and colour-matched. Measured: eye sharpness 13 → 25 (the photo is 26), mouth 8 → 16. Optional GFPGAN "Sharpen the face" restores every frame (much slower). Clips are cached, so a repeated answer plays instantly.

The greeting

When the app opens, the avatar says hello in the person's voice ("Hi, I'm X's digital twin…"). It is made once through the same voice and lip-sync queue, then served from the cache.

15 · The landing character

The home page shows an animated cartoon character whose head and eyes follow the mouse. It is not a 3D model; it is a **grid of pre-rendered images**:

- The character image was cleaned with **Real-ESRGAN**, then **LivePortrait** (a face re-animation model) rendered it at yaw −20°…+20° in 1° steps and pitch −10°…+10° in 2.5° steps (369 poses), plus half-closed and closed eyes for blinking (90).
- Each frame was colour-matched and detail-restored with optical flow, the background removed, and saved as WebP.
- **Eye layer**: the irises were painted out of every frame and are drawn live by the page, clipped to each eye opening and moved toward the cursor, so the eyes really follow you.
- **Smooth motion**: a critically damped spring moves the head (settles in ~150 ms); every animation frame blends the 4 nearest poses; images are decoded by background Web Workers so the page never stutters; small always-ready copies stand in during fast moves.
- Measured in the app's own WebKit engine: 60 frames per second, no wrong-pose frames while moving.

16 · The extra features

Feature	What the person sees	How it works
Why this answer	Under every answer: the real answers it used, each with a similarity bar and the 0.15 / 0.35 marks	Shows the retrieved evidence and raw scores; marks which ones were above the floor ("used").
What would I do?	Describe a choice + options; the twin picks one with reasons and certainty	Retrieval on the situation gives the confidence (below 0.15 → "I don't know"); Gemini or Phi-3 picks an option citing your answers; never "high" certainty below 0.35; each option's closeness to your answers is shown.
Guess me	A new question; the twin's guess is locked before you answer; then a match score	The guess is kept on the server (can't be peeked); similarity of meaning is the score; your answer becomes memory, so the score can climb.
Personality snapshot	A Big Five bar chart on the Teach page	The 20 "How you tend to be" items: 4 per trait, some reverse-scored (e.g. "Long social events leave me drained" counts against Extraversion), averaged and scaled 0–100.
Change photo / voice	Settings: replace either at any time	Same checks as the interview; every cached voice/video clip is cleared so they're made again.

17 · How the twin is measured

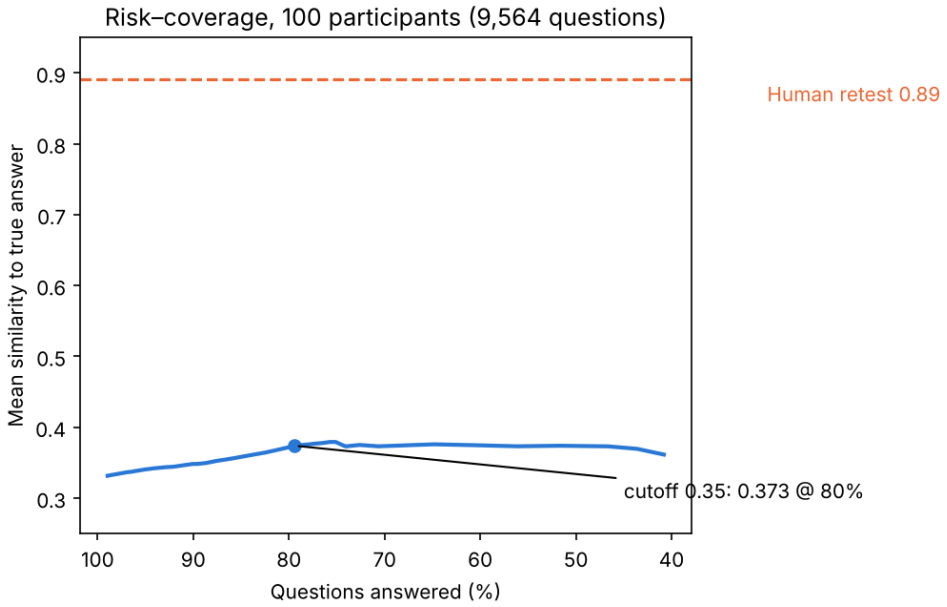
The key idea: **hide some real answers, ask the twin those questions, and compare**. The twin never saw the hidden answers, so this tests whether it can predict the person, not whether it remembers.

- **Score**: semantic similarity (cosine of embeddings) between the twin's answer and the real answer, from 0 to 1. This captures meaning, so "I'd fix it myself" and "I'd repair it on my own" score high.
- **Human ceiling**: people don't answer identically twice. The same person answering a reworded question again gives the best realistic score. A twin can't be expected to match you better than you match yourself.
- **Per-tier scores**: how good answers are in each tier, and how often each tier is used (coverage).
- **Selective accuracy**: if the twin only answered when confident, how good would it be, and how much would it cover?

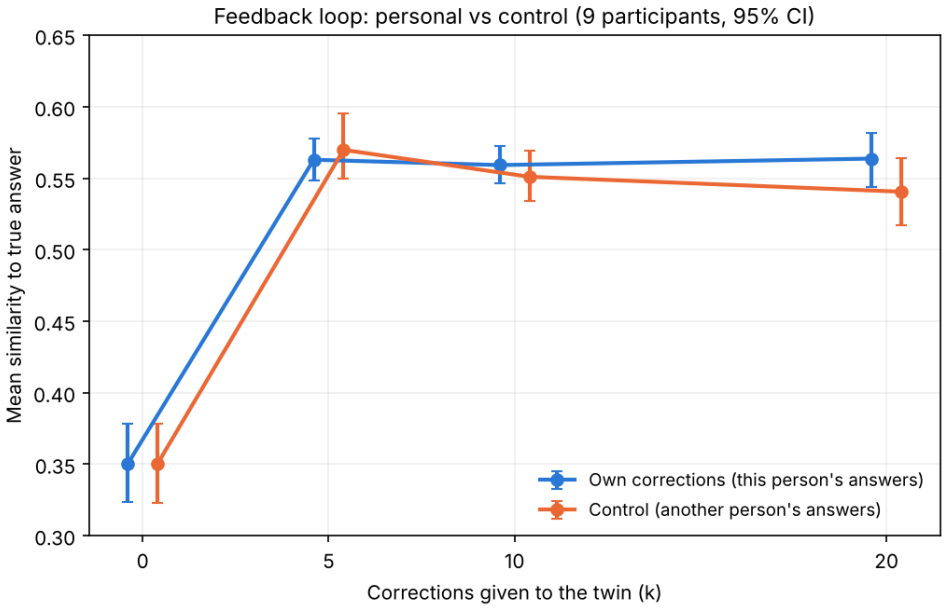
18 · Phase 1: the public benchmark (100 people)

Phase 1 used **Twin-2K-500**, a public research dataset of survey answers from real people. 100 participants, **9,564 held-out questions**. (These results are frozen and reproducible: the evaluation script and data are never modified.)

Measure	Value	Meaning
Twin similarity	0.332 (95% CI 0.325–0.339)	average closeness of the twin's answers to the true ones
Human re-test ceiling	0.892	the same people answering again (multiple-choice survey items are very repeatable)
Confident answers only	0.373 at 80% coverage	+0.042 better when it only answers when confident: the gate selects better answers
Tier shares	confident 79.6% · guess 19.4% · refuse 1.0%	guesses score 0.16 vs 0.373 for confident: the tiers sort quality correctly
Correction loop (9 people)	0.35 → 0.563 after 5 corrections	learning without training works; a control reached 0.57; 20 corrections on personal facts +0.023



Risk–coverage: answering only above a confidence level raises quality; lower coverage, higher accuracy.



Feedback curve: similarity on related questions rises with each correction, without training.

19 · Phase 2: your own twin (the 18 questions)

Phase 2 tests the twin of the real subject (Hardhik). From the interview, a stratified 25% of answers was hidden: **18 questions** the twin never saw. Each was asked to the twin, and its answer compared with the real one.

Measure	Value	In plain words
Twin average	0.57	on questions it never saw, the twin's answers are fairly close in meaning to Hardhik's
Your own ceiling	0.79	Hardhik vs Hardhik: the "Quick re-check" reworded repeats match the originals 0.79
Of ceiling	72%	0.57 / 0.79: the twin is about 72% as consistent with Hardhik as Hardhik is with himself
Confident answers	12 of 18 (67%)	average 0.48
Guessed answers	6 of 18 (33%)	average 0.77 (see below)
Refused	0	every hidden question was related to something he said

> **One-line answer for the panel:** "On 18 questions it had never seen, my twin answered about 72% as consistently as I answer myself, and on the public benchmark of 100 people the confidence gate reliably separated good answers from guesses."

Why do guesses score higher than confident answers here?

With only 18 questions, a few answers swing the averages (6 guesses vs 12 confident). Several hidden questions were either-or or rating questions where a profile-based guess happens to be right, while some confident answers were long free-text answers that score lower on similarity even when correct. On the 9,564-question Phase 1 benchmark the expected order holds clearly (confident 0.373 vs guess 0.16). **Be honest about the small sample**: it is one person's interview.

The Results tab has a "Score the twin now" button that re-runs this evaluation with all current corrections included.

20 · Technology stack

Area	Technology	Why
App shell	pywebview + macOS WebKit; custom .app bundle with its own Python, ad-hoc signed, plus a .dmg	native window, no browser needed
Server	Python 3.11, FastAPI, Uvicorn	simple JSON API, streaming answers
Front end	Vanilla JavaScript, CSS, Canvas, Web Workers, SVG	no framework; fast and small
Language model	Phi-3-mini-4k-instruct, 4-bit, Apple MLX (mlx-lm)	runs on a 16 GB Mac in ~2.4 GB, ~1 s/answer
Embeddings	sentence-transformers all-mpnet-base-v2 (768-d)	strong semantic similarity
Vector search	FAISS IndexFlatIP (exact cosine)	fast, exact for a few hundred facts
Online model	Google Gemini via google-genai SDK; key in Keychain via keyring	optional better reasoning
Voice	Coqui XTTS-v2	zero-shot voice cloning, multilingual incl. Hindi
Translation	Helsinki-NLP opus-mt (Marian) en↔hi; Gemini online	offline Hindi
Speech recognition	OpenAI Whisper-small (transformers)	multilingual, local
Lip-sync	Wav2Lip (GAN) + OpenCV mouth-only blend; optional GFPGAN	one photo → talking video
Character	LivePortrait, Real-ESRGAN, OpenCV optical flow	pre-rendered head-turn grid
Packaging	uv with locked requirements; bootstrapper downloads models	same versions on every Mac
Tests	6 test files with stand-in models (no network)	gate honesty, key safety, every feature

21 · File map: which file does what

File	Role
app/server.py	The API: ask, build, correct, notes, media, online, expert, decide, Big Five, Guess me, talk
app/static/app.js, styles.css	Every page, the chat, the character player, charts
app/online.py	Gemini: Keychain key, model chain, profile, prompts, fallback
app/work.py	Expert mode: job profile, field subjects, field-question matching
app/lang.py	Language detection, translation, follow-up rewriting
app/asr.py	Whisper speech recognition (Talk mode)
app/media.py	Queue for voice and video jobs, clip cache
src/rag_pipeline.py	Facts, embeddings, FAISS index, retrieve() with raw confidence
src/deferral.py	The three tiers, prompts, answer formats, cleaning, corrections, notes
src/persona.py	Profile and style card from the interview
src/llm_backend.py	Loads Phi-3 on MLX (or a stand-in for tests)
src/multimodal.py	XTTS voice, Hindi translation, Wav2Lip + mouth blend, GFPGAN
src/load_phase2.py, eval_phase2.py	Build the twin from the interview; Phase 2 evaluation
src/eval_benchmark.py	Phase 1 evaluation (frozen)
bootstrap/engine.py, launcher.py	First-run setup and launching the window
tools/liveportrait/*	Rendering and processing the landing character
tests/*.py	Automated checks for every feature

22 · Privacy, ethics and the project's hard rules

- **Local by default**: answers, memory, voice and photo stay in the app folder on the Mac. The server only accepts local connections.
- **Online mode is opt-in**, with its own consent line; voice and video never leave the Mac; the key is in the Keychain only.
- **Consent boundary**: voice cloning and the avatar only for the person who consented; benchmark participants (anonymous) never get a voice or face.
- **No training**: the language model is frozen; personal data is never baked into model weights.
- **Honest deferral**: the gate uses raw similarity; the boost never reaches it; there is a test.
- **Frozen evaluation**: the Phase 1 script and data are never modified, so the published numbers stay reproducible.
- **Delete anytime**: Settings can delete the twin, or remove everything including models.

23 · Limitations and future work

Limitations (say these before they are asked)

- **Small personal test set**: 18 hidden questions for one person; averages move with a few answers.
- **Similarity is a proxy**: cosine similarity measures closeness of meaning, not correctness; two different but valid answers can score low.
- **Fixed thresholds**: 0.15 and 0.35 were tuned on a survey dataset; the best values may differ per person.
- **Speech**: Whisper-small is weak in Telugu; the cloned Hindi voice has an English accent (the voice sample is English); no Telugu voice.
- **Avatar**: Wav2Lip moves only the mouth and is slow on a Mac (minutes per answer); the head is still.
- **Expert mode** depends on Gemini and on correctly recognising field questions.

Future work

- Per-person calibration of the thresholds from their own corrections.
- An Indic voice model (e.g. AI4Bharat) for natural Hindi and a Telugu voice; a larger Whisper for Telugu.
- A newer talking-head model with head motion and expressions.
- Asking the person the questions it is least sure about (active learning).
- A Windows version (CUDA instead of MLX) and secure sharing of a twin with family.

24 · Questions examiners are likely to ask

Q. How is this different from ChatGPT with a prompt?

Three things: it refuses when nothing related is known (no model call below 0.15, so no invention), every answer shows the person's real answers it came from, and it learns from corrections instantly without training. It also runs offline on a laptop.

Q. Did you train or fine-tune a model?

No. The model is frozen. The twin learns by adding facts to a retrieval index; that's why a correction works on the next question.

Q. What is the confidence, exactly?

The highest cosine similarity between the question's embedding and any of the person's fact embeddings (all-mpnet-base-v2, FAISS), using the raw value, not the boosted ranking score.

Q. Why 0.15 and 0.35?

Chosen on the Phase 1 benchmark from the risk–coverage curve: below 0.15 almost nothing relevant exists; above 0.35 answers are reliably grounded. They are fixed for all features.

Q. Can adding notes fake confidence?

No. Notes and corrections get a +0.05 boost only for ranking. The gate uses raw similarity, and a test enforces this.

Q. What does 72% mean?

On 18 hidden questions the twin scored 0.57 similarity; the person matches their own earlier answers at 0.79. 0.57/0.79 = 72% of the realistic maximum.

Q. Why is the Phase 1 twin score (0.332) lower than Phase 2 (0.57)?

Different data: Phase 1 is a large survey dataset with many multiple-choice items and a very high human re-test (0.892); Phase 2 is one person's open interview. Compare each to its own ceiling, not to each other.

Q. Where does the persona/style come from?

From the interview: persona.py computes typical answer length, lowercase use, rating habits, Big Five and example answers, and puts them in the prompt.

Q. Is the voice cloning trained on my voice?

No. XTTS-v2 is zero-shot: it conditions on a 28-second sample at generation time.

Q. How does the lip-sync work?

Wav2Lip predicts mouth shapes from the audio for each frame of the photo; the app keeps the original photo and blends only the mouth region, sharpened and colour-matched.

Q. What happens if Gemini is down?

The model chain tries other Gemini models; if all fail, the local Phi-3 answers and the card says "offline · this mac" with the reason.

Q. Is Online mode safe for privacy?

It is off by default and needs consent. Only text is sent (profile, the question, closest answers); voice and video stay local; the key is in the Keychain.

Q. How do you stop Expert mode from making things up about the person?

It only triggers for field questions when personal memory has nothing confident, is labelled "Expert knowledge", and the prompt forbids personal claims; personal questions still go through the gate.

Q. Does the chat use all interview questions?

Its memory is your interview answers (except the 18 held back for testing), plus every correction and note. Online, Gemini also reads a profile written from them, and Expert mode adds general field knowledge, clearly labelled.

Q. What would you improve with more time?

Per-person threshold calibration, a better Indic voice, a head-moving avatar, and active learning (asking the questions it's least sure about).

25 · Glossary

Term	Meaning
Digital twin	Software that answers as a specific person would
Deferral	Choosing not to answer ("I don't know") when evidence is too weak
Embedding	A list of numbers (here 768) representing the meaning of a text
Cosine similarity	How closely two embeddings point the same way: 1 = same meaning, 0 = unrelated
RAG	Retrieval-Augmented Generation: find relevant facts, then let the model answer from them
FAISS	A library for fast vector search (Facebook AI Similarity Search)
LLM	Large language model (here Phi-3-mini locally, Gemini online)
Quantisation (4-bit)	Storing model weights with fewer bits so the model fits in less memory
MLX	Apple's machine-learning framework for Apple-silicon GPUs
Fine-tuning	Training a model further on new data (not used here)
Held-out set	Answers hidden from the twin, used only for testing
Human ceiling	How well the person matches themselves when asked again
Likert scale	1–5 agreement scale (Strongly disagree … Strongly agree)
Big Five	Openness, Conscientiousness, Extraversion, Agreeableness, Neuroticism
XTTS-v2	A text-to-speech model that clones a voice from a short sample
Wav2Lip	A model that makes a face's mouth move in sync with audio
Whisper	OpenAI's speech-recognition model
Coverage	The share of questions the twin answers (instead of refusing)